Functions & Graphs

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

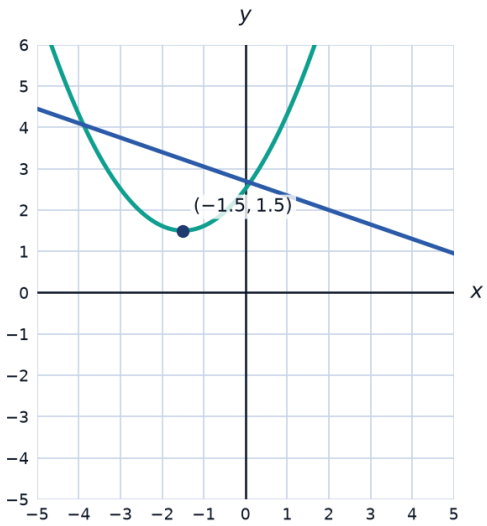
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A car travels 120 miles in 3 hours. Find the average speed.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the least common multiple of 6 and 8.



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Give the equation of the horizontal line through $(0, 5)$.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the distance between $(0, 0)$ and $(3, 4)$.

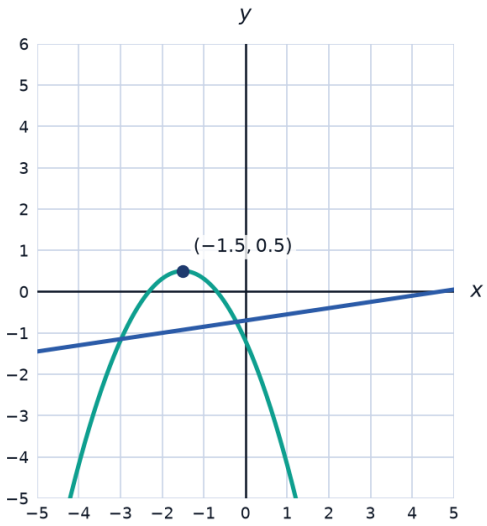
5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the interior-angle sum of a pentagon.

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6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For what value of x is $\frac{1}{x-2}$ undefined?



7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A right triangle has opposite 6 and adjacent 8. Find $\tan\theta$.

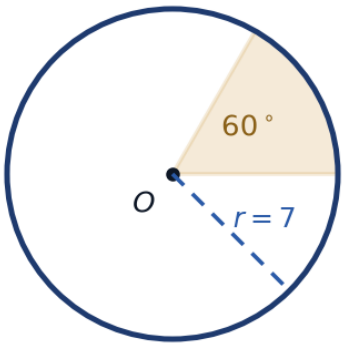
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the midpoint of $(2, 4)$ and $(6, 8)$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

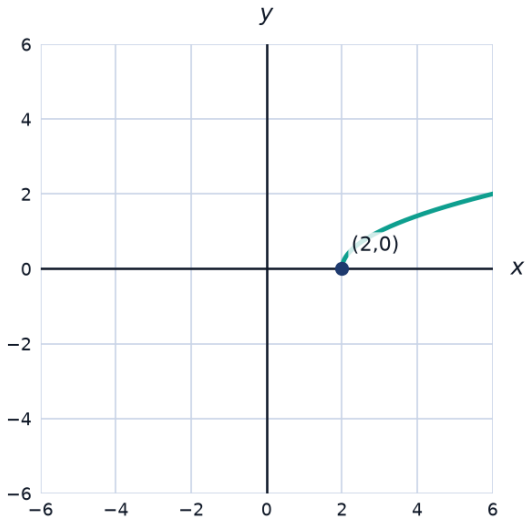
Find the circumference of a circle with $r = 7$. Take $\pi = \frac{22}{7}$.



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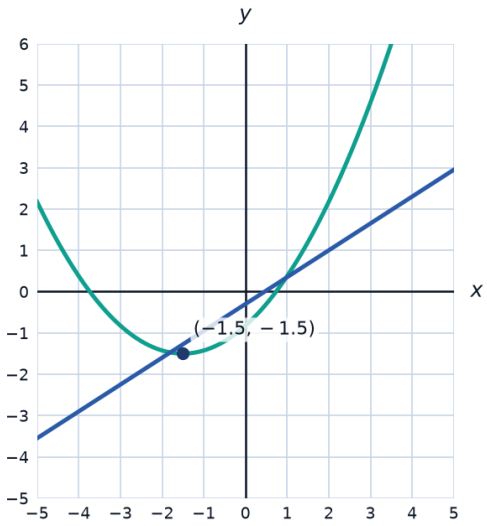
10 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

What is the smallest value of x in the domain of $f(x) = \sqrt{x-2}$? Show your complete algebraic work.



11 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

The graph of $y = -f(x)$ is the graph of $y = f(x)$ transformed how? Show your complete algebraic work.



12 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

The graph of $y = f(x-5)$ is the graph of $y = f(x)$ shifted how? Show your complete algebraic work.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Practice 3.2: What is the domain of the function $f(x) = \frac{\sqrt{x+5}}{x-3}$?

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

What is the inverse function $f^{-1}(x)$ of $f(x) = \frac{5x+2}{3}$?

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Coordinate Geometry

Level 3 · AdvancedIn the hard Module 2, questions 18–22 are typically high-difficulty multi-step problems with parameters k. If a student has 7 minutes remaining for the last 3 questions, which approach optimizes score yield: attempting all 3 partially, or using Desmos regression and choice elimination to guarantee complete accuracy on 2 of them?

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Plane Geometry

On the enhanced ACT Math test (45 questions in 50 minutes), a student allocates 10 minutes to the first 15 questions and 16 minutes to questions 16 through 30. How many seconds per question remain for the final 15 questions, assuming 3 minutes are reserved at the end to check answers?

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Trigonometry

Practice 1.2: If a and b are positive integers such that the greatest common factor of a and b is 6, and the least common multiple of a and b is 72, what is the product $a \times b$?

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Statistics & Probability

Why should you test Choice C (or H) first when back-solving numerical answer choices on the ACT?

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

What is the greatest common factor (GCF) and least common multiple (LCM) of 24 and 36?

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

For what values of x does $|2x - 5| > 9$ hold?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given the linear function $f(x) = \frac{4x - 7}{3}$, what is the equation of the inverse function $f^{-1}(x)$?

a. $f^{-1}(x) = \frac{3x + 7}{4}$

b. $f^{-1}(x) = \frac{3x - 7}{4}$

c. $f^{-1}(x) = \frac{4x + 3}{7}$

d. $f^{-1}(x) = \frac{7x + 3}{4}$

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $g(x) = \sqrt{2x + 10}$, what is the domain of $g(x)$ in the set of real numbers?

a. $x \geq -5$

b. $x > -5$

c. $x \geq 0$

d. $x \leq -5$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let $f(x) = 2x + 5$ and $g(x) = x^2 - 3$. What is the value of $(g \circ f)(1)$?

a. 11

b. 46

c. 49

d. 52

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $h(x) = 3x - 1$ and $(h \circ k)(x) = 6x + 8$, which of the following defines $k(x)$?

a. $k(x) = 2x + 3$

b. $k(x) = 2x + 9$

c. $k(x) = 3x + 4$

d. $k(x) = 2x + 2$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the range of the quadratic function $f(x) = -(x - 4)^2 + 9$?

a. $y \leq 9$

b. $y \geq 9$

c. $y \leq 4$

d. All real numbers

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The graph of $y = f(x)$ is shifted 3 units to the right and 5 units down. What is the equation of the transformed graph?

a. $y = f(x - 3) - 5$

b. $y = f(x + 3) - 5$

c. $y = f(x - 3) + 5$

d. $y = f(x + 3) + 5$

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which of the following functions is an even function, satisfying $f(-x) = f(x)$ for all real x?

a. $f(x) = x^3 - x$

b. $f(x) = 4x^2 + 7$

c. $f(x) = 2x + 1$

d. $f(x) = x^4 - 3x$

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the horizontal asymptote of the rational function $f(x) = \frac{6x^2 - 5x + 1}{2x^2 + 3x - 4}$?

a. $y = 0$

b. $y = 3$

c. $y = 2$

d. No horizontal asymptote

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A function is defined piecewise by $f(x) = 2x + 3$ for $x < 1$, and $f(x) = x^2 - 1$ for $x \geq 1$. What is the value of $f(-2) + f(3)$?

a. 7

b. 8

c. 9

d. 11

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If $[x]$ denotes the greatest integer function (floor function), what is the value of $[3.8] - [-2.4]$?

a. 5

b. 6

c. 7

d. 1

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For what value of c is the piecewise function $f(x) = cx + 4$ for $x \leq 2$, and $f(x) = 3x^2 - 2$ for $x > 2$ continuous at $x = 2$?

a. 2

b. 3

c. 4

d. 5

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

What is the domain restriction for $f(x) = \frac{x+1}{x-9}$? (Give the excluded x-value.)

a. 11

b. 14

c. 3

d. 9

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

What value of x is excluded from the domain of $f(x) = \frac{1}{x}$?

a. -1

b. 1

c. 5

d. 0

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In the standard (x, y) coordinate plane, what is the area, in square coordinate units, of the region enclosed by the graphs of the functions $f(x) = |x - 4| - 3$ and $g(x) = 5 - |x - 4|$?

a. 16

b. 32

c. 48

d. 64

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For $y = (x + 3)^2 + 2$, what is the axis of symmetry?

a. x = 0

b. x = -5

c. x = -3

d. x = -1

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $f(x) = x - 3$, find the average rate of change from $x = 1$ to $x = 3$.

a. 3

b. -3

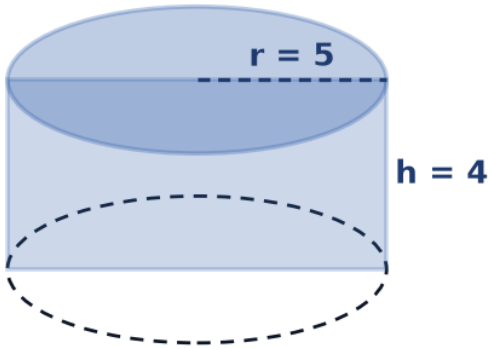
c. 5

d. 1

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A cylinder has radius 5 and height 4. What is its volume (in terms of π)?



a. 144π

b. 100π

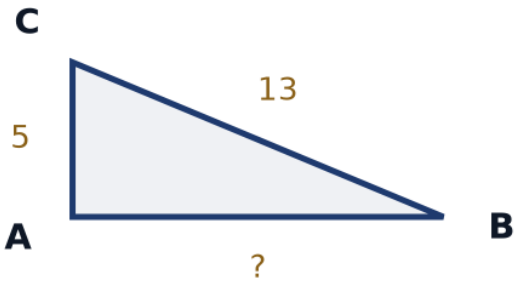
c. 40π

d. 20π

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Statistics & Probability

A right triangle has hypotenuse 13 and one leg 5. What is the length of the other leg?



a. 28

b. 33

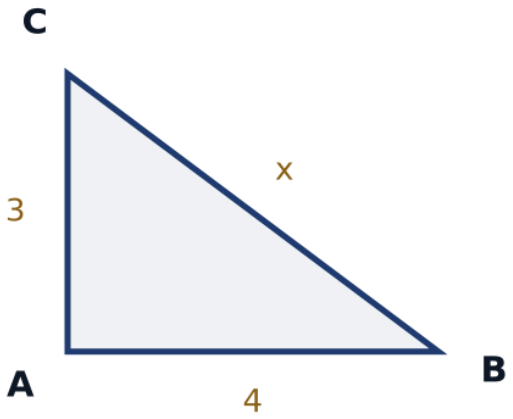
c. 22

d. 12

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

A right triangle has legs 3 and 4. What is the length of the hypotenuse?



a. 7

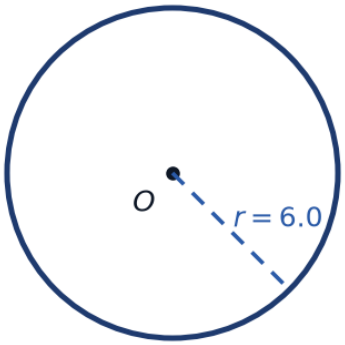
b. 5

c. 29

d. 13

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

A circle has radius 6. What is its area (in terms of π)?



a. 49π

b. 36π

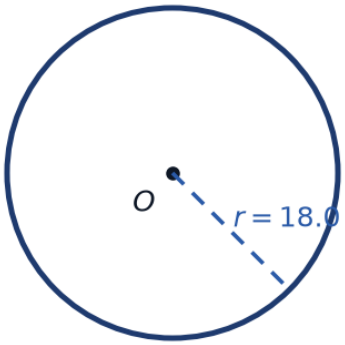
c. 12π

d. 72π

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

A circle has diameter 18. What is its area (in terms of π)?



a. 162π

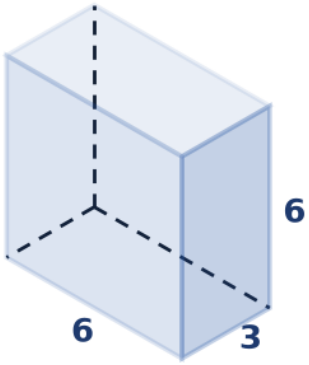
b. 18π

c. 100π

d. 81π

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Coordinate Geometry

A rectangular prism has dimensions $6 \times 3 \times 6$. What is its volume?



a. 203

b. 61

c. 235

d. 108

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Plane Geometry

An isosceles trapezoid has parallel bases of lengths 4 cm and 16 cm. A circle is inscribed inside the trapezoid such that it is tangent to all four sides. What is the area, in square centimeters, of the region inside the trapezoid that lies outside the circle?

a. $80 - 16\pi$

b. $80 - 25\pi$

c. $100 - 25\pi$

d. $64 - 16\pi$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

A box contains 10 cards, each labeled with a distinct integer from 1 through 10. If 3 cards are drawn at random from the box without replacement, what is the probability that the sum of the numbers on the 3 drawn cards is divisible by 3?

a. $\frac{1}{4}$

b. $\frac{3}{10}$

c. $\frac{1}{3}$

d. $\frac{7}{20}$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Statistics & Probability

In the standard (x, y) coordinate plane, a triangle has vertices at $O(0, 0)$, $A(8, 0)$, and $B(2, 6)$. A horizontal line with equation $y = c$, where $0 < c < 6$, divides the interior of ΔOAB into two regions of equal area. What is the value of c?

a. $6 - 3\sqrt{2}$

b. 3

c. $3\sqrt{2}$

d. $6 - 2\sqrt{3}$

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